1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In non-small cell lung cancer (NSCLC), loss of E-cadherin and the broader EMT program (EMT-high state) reprogram the tumor immune microenvironment such that high tumor-infiltrating lymphocyte (TIL) density reflects a predominantly peritumoral or dysfunctional T cell population with impaired cytotoxicity, rather than a productive intratumoral anti-tumor response. Consequently, EMT-high tumors with high TIL density will exhibit attenuated benefit from pembrolizumab monotherapy compared with EMT-low tumors with similar TIL densities, despite comparable PD-L1 status or tumor mutational burden. Mechanistically, EMT fosters T cell exclusion and functional impairment through: (a) ECM/fibroblast-rich stroma and altered chemokine landscapes favoring suppressive myeloid cells and regulatory T cells; (b) upregulation of inhibitory pathways and exhaustion markers on intratumoral T cells; (c) potential modulation of PD-L1 expression via EMT-associated signaling (e.g., TGF-β, ZEB1) but with the dominant effect being spatial and functional impairment of T cells rather than PD-L1 abundance alone. This EMT-TIL interaction should be detectable by (i) EMT status (E-cadherin loss and EMT gene signature) correlating with TIL localization (intratumoral vs peritumoral) and functional status (GZMB, IFN-γ, PD-1, TIM-3), (ii) imaging features of lesion morphology and invasion context (ground-glass nodules, CIS) providing contextual timing of EMT development, and (iii) clinical outcomes to pembrolizumab. The working prediction: TIL density is a positive predictor of pembrolizumab response only in EMT-low tumors; in EMT-high tumors, high TIL density does not translate to improved response or longer event-free survival.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	E-cadherin, EMT status
Primary endpoint	therapy_response (pembrolizumab monotherapy)

2. Why This Gene — Not Others

Primary hypothesis gene(s)

E-cadherin: E-cadherin) should alter the tumor-immune interface, yet TIL-rich tumors do not uniformly benefit from PD-1 blockade—why does immune infiltration sometimes fail to translate into pembrolizumab response? 2) Built on Tier 1 paths showing (a) EMT status in lung cancer links to pembrolizumab benefit (Core Path 1) and (b) E-cadherin loss is a practical EMT readout with prognostic value (Core Path 2).

EMT status: EMT status in lung cancer links to pembrolizumab benefit (Core Path 1) and (b) E-cadherin loss is a practical EMT readout with prognostic value (Core Path 2).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	PD-L1	Supporting modifier from knowledge graph
Secondary	ground-glass nodules (GGN)	Supporting modifier from knowledge graph
Secondary	CIS	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective NSCLC cohorts treated with pembrolizumab with available tumor tissue; quantify EMT status via E-cadherin IHC and EMT gene expression, quantify TIL density and spatial localization by multiplex IHC/IF, assess T cell functionality (GZMB, IFN-γ, PD-1, TIM-3); perform multivariable analyses with PD-L1 and TMB as covariates and test TIL-EMT interaction.

Prediction 2: Spatial transcriptomics to map chemokine/ECM programs and immune cell states relative to EMT status.

Prediction 3: Preclinical validation using NSCLC cell lines/ORGs with E-cadherin knockdown/overexpression co-cultured with cytotoxic T cells and/or treated with PD-1 blockade; in vivo models with EMT induction/inhibition to assess TIL localization and response.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.
- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PD-L1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- ground-glass nodules (GGN): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.